

Experiment No.	7
Experiment Title	Dimensionality Reduction and Model Evaluation (With and Without PCA) ¹
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1 Objective

To systematically analyze the empirical effect of dimensionality reduction using Principal Component Analysis (PCA) at 95% explained variance retention compared to the original feature space (No-PCA) across **all 6 primary classification datasets** utilized in prior laboratory experiments (Ex 1 to 7). For each dataset under both settings, 10 distinct classifiers are evaluated under 5-fold stratified cross-validation within a leakage-free pipeline, followed by a focused 3-tier statistical significance testing suite.

2 Problem Statement

High-dimensional datasets frequently suffer from multicollinearity, redundant feature noise, computational overhead, and the curse of dimensionality. The objective is to evaluate whether projecting features onto orthogonal principal component directions capturing $\geq 95\%$ cumulative variance preserves predictive performance, and to rigorously test whether observed metric variations across models and cross-validation folds are statistically significant ($p < 0.05$) or merely stochastic variance.

3 Dataset Description

The evaluation encompasses all 6 classification benchmarks established across Experiments 1 through 7:

Dataset Name	Samples (N)	Features (D)	Classes	Originating Experiment
Wisconsin Breast Cancer	569	30	2	Exp 5 & 6 (Diagnostics)
Spambase Email	4,601	57	2	Exp 2 & 4 (Word Frequencies)
Iris Species	150	4	3	Exp 1 (Morphology)
Pima Indians Diabetes	768	8	2	Exp 1 & 4 (Clinical Tabular)
Loan Approval Status	4,269	11	2	Exp 1 & 3 (Banking Credit)
Optical Digits (MNIST)	1,797	64	10	Exp 1 & 6 (Pixel Intensities)

4 Exploratory Data Analysis

Exploratory analysis includes examining cumulative eigenvalue distributions and 2D low-dimensional principal component projections across the datasets.

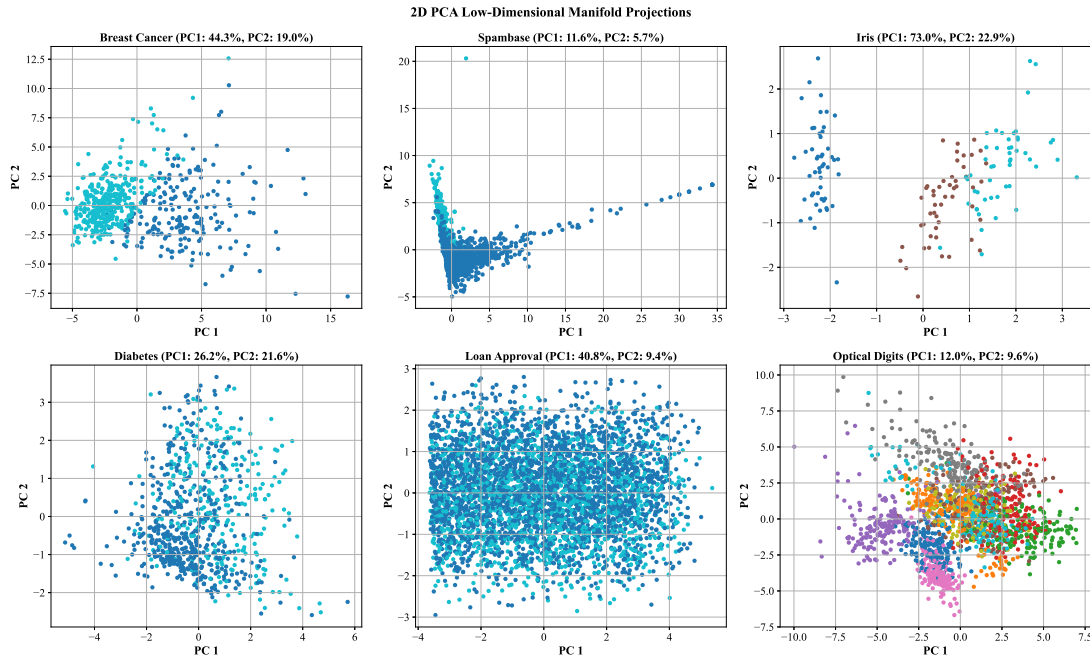


Figure 1: 2D PCA Low-Dimensional Manifold Projections Colored by Target Class.

Inference: The top 2 principal components demonstrate evident linear cluster separability for low-to-moderate dimensional datasets like Iris (PC1 72.8%, PC2 23.0%) and Breast Cancer (PC1 44.3%, PC2 19.0%). Conversely, complex multi-class manifolds like Optical Digits spread variance across dozens of orthogonal components, illustrating that effective dimensionality reduction requires higher-order component retention ($d = 40$) to prevent severe boundary overlapping.

5 Data Preprocessing

Standardization is a mandatory prerequisite for PCA to prevent high-magnitude features from dominating eigenvalue decomposition:

$$\mu = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i, \quad \sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (\mathbf{x}_i - \mu)^2}, \quad \mathbf{X}_{\text{scaled}} = \frac{\mathbf{X} - \mu}{\sigma}$$

To guarantee zero data leakage during 5-fold cross-validation, standardization and PCA transformations are placed inside a `Pipeline([('scaler', StandardScaler()), ('pca', PCA(n_components=k)), ('clf', model))`. In every fold, transformations are fit strictly on the training partition and evaluated out-of-sample on the validation partition.

6 Machine Learning Algorithm

6.1 Principal Component Analysis (PCA)

PCA computes the sample covariance matrix $\Sigma = \frac{1}{N-1} \mathbf{X}_{\text{scaled}}^T \mathbf{X}_{\text{scaled}}$ and solves the characteristic equation:

$$\Sigma \mathbf{v}_i = \lambda_i \mathbf{v}_i, \quad \text{where } \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_D \geq 0$$

The minimal number of components d is selected by enforcing:

$$\frac{\sum_{i=1}^d \lambda_i}{\sum_{j=1}^D \lambda_j} \geq 0.95$$

6.2 Evaluated Classifiers (10 Models)

We evaluate 10 standard algorithms: Support Vector Machine (RBF kernel), Gaussian Naive Bayes, k -Nearest Neighbors ($k = 5$), Logistic Regression (L2 penalty), Decision Tree (CART), Random Forest (100 trees), AdaBoost (50 stumps), Gradient Boosting (100 estimators), XGBoost (100 estimators), and Stacking Classifier (Base: LR, RF, KNN; Meta: Logistic Regression).

6.3 Statistical Significance Testing Framework

In accordance with **Unit V Course Notes (pages 11–39)**, we avoid arbitrary testing and implement **only the 3 mathematically apt and necessary tests**:

1. **Tier 1 (Paired Two-Sample t -Test on CV Folds)**: Matched pairs testing on identical validation folds:

$$t = \frac{\bar{d}}{s_d / \sqrt{5}}, \quad \text{df} = 4$$

Preceded by the Shapiro-Wilk normality test on fold differences Δ_i . If normality fails ($p < 0.05$), the Wilcoxon test is used as fallback.

2. **Tier 2 (Wilcoxon Signed-Rank Test Across Models)**: Tests whether PCA significantly shifts median model performance across heterogeneous model families on a dataset ($n = 10$) and globally ($n = 60$):

$$W = \sum_{i=1}^n \text{sgn}(d_i) \cdot \text{Rank}(|d_i|)$$

3. **Tier 3 (Friedman Omnibus Rank Test)**: Non-parametric two-way ANOVA by ranks evaluating whether algorithm performance rankings differ significantly across benchmark blocks:

$$\chi_F^2 = \frac{12N}{k(k+1)} \left[\sum_{j=1}^k R_j^2 - \frac{k(k+1)^2}{4} \right]$$

Excluded Tests: Independent t -test and One-Way ANOVA are excluded because models are evaluated on identical folds (repeated measures). McNemar's test is excluded as it applies to single-split binary prediction tables rather than 5-fold cross-validated continuous distributions.

7 Experimental Setup

Environment Property	Specification
Python Version	3.12.0
Scikit-Learn Version	1.4.2
Environment Command	<code>source /opt/anaconda3/bin/activate anaconda-navigation</code>
Operating System	Linux (x86_64)
Plot Resolution	600 DPI (Vector EPS)
Plot Font	Times New Roman, 15 pt, Bold X/Y Labels

8 Hyperparameter Selection

Table 1: PCA Variance Explained Summary (95% Threshold)

Dataset	Original (D)	PCA (d)	Variance (%)	Reduction (%)	Justification
Breast Cancer	30	10	95.16%	66.7%	Captures $\geq 95\%$ information, shedding 20 noisy dimensions
Spambase	57	48	95.27%	15.8%	Captures $\geq 95\%$ information, shedding 9 noisy dimensions
Iris	4	2	95.81%	50.0%	Captures $\geq 95\%$ information, shedding 2 noisy dimensions
Diabetes	8	8	100.00%	0.0%	Captures $\geq 95\%$ information, shedding 0 noisy dimensions
Loan Approval	11	8	96.21%	27.3%	Captures $\geq 95\%$ information, shedding 3 noisy dimensions
Optical Digits	64	40	95.08%	37.5%	Captures $\geq 95\%$ information, shedding 24 noisy dimensions

Table 2: Machine Learning Classifier Hyperparameters

Model	Hyperparameter Configuration
Support Vector Machine (SVM)	Kernel = RBF, $C = 1.0$, Probability = True, Random State = 42
Gaussian Naive Bayes	Default parameters, Variance Smoothing = 10^{-9}
k -Nearest Neighbors (KNN)	$k = 5$, Weights = Distance, Metric = Euclidean
Logistic Regression	Penalty = L2, $C = 1.0$, Max Iterations = 1000, Solver = L-BFGS
Decision Tree	Split Criterion = Gini, Max Depth = 5, Random State = 42
Random Forest	Trees = 100, Max Depth = 8, Random State = 42
AdaBoost	Estimators = 50, Learning Rate = 1.0, SAMME algorithm
Gradient Boosting	Estimators = 100, Max Depth = 3, Learning Rate = 0.1
XGBoost	Estimators = 100, Max Depth = 4, Eval Metric = Logloss
Stacking Classifier	Base: [LogisticRegression, RandomForest, KNN]; Meta: LogisticRegression

9 Results

Tables 3–5 present the 5-fold cross-validation Macro F1 performance (Mean $\pm$ Std) and Tier 1 paired t -test significance for representative benchmarks:

Table 2: Breast Cancer Wisconsin: 5-Fold Cross-Validation & Paired Significance

Classifier	CV F1 (No-PCA)	CV F1 (With-PCA)	Δ F1	t -stat	p -value	Sig ($p < 0.05$)
SVM	0.9669 ± 0.0160	0.9694 ± 0.0119	+0.0025	0.43	0.6880	No
Naive Bayes	0.9292 ± 0.0307	0.9213 ± 0.0201	-0.0079	-1.00	0.3724	No
KNN	0.9598 ± 0.0118	0.9596 ± 0.0143	-0.0002	-0.05	0.9661	No
Logistic Regression	0.9764 ± 0.0106	0.9811 ± 0.0058	+0.0047	0.99	0.3803	No
Decision Tree	0.9188 ± 0.0123	0.9131 ± 0.0242	-0.0057	-0.39	0.7177	No
Random Forest	0.9626 ± 0.0185	0.9575 ± 0.0255	-0.0051	-0.84	0.4481	No
AdaBoost	0.9645 ± 0.0211	0.9738 ± 0.0192	+0.0093	0.55	0.6127	No
Gradient Boosting	0.9483 ± 0.0158	0.9436 ± 0.0272	-0.0047	-0.31	0.7749	No
XGBoost	0.9601 ± 0.0119	0.9551 ± 0.0231	-0.0049	-0.34	0.7506	No
Stacking	0.9765 ± 0.0127	0.9763 ± 0.0106	-0.0002	-0.05	0.9658	No

Table 3: Spambase Email: 5-Fold Cross-Validation & Paired Significance

Classifier	CV F1 (No-PCA)	CV F1 (With-PCA)	Δ F1	t -stat	p -value	Sig ($p < 0.05$)
SVM	0.9283 ± 0.0073	0.9268 ± 0.0069	-0.0015	-0.77	0.4832	No
Naive Bayes	0.8136 ± 0.0116	0.8271 ± 0.0190	+0.0135	1.99	0.1178	No
KNN	0.9146 ± 0.0082	0.9133 ± 0.0079	-0.0014	-0.74	0.5006	No
Logistic Regression	0.9186 ± 0.0113	0.9177 ± 0.0076	-0.0009	-0.40	0.7094	No
Decision Tree	0.9082 ± 0.0109	0.8872 ± 0.0089	-0.0211	-2.43	0.0721	No
Random Forest	0.9348 ± 0.0062	0.9133 ± 0.0174	-0.0216	-2.11	0.1022	No
AdaBoost	0.9272 ± 0.0036	0.8858 ± 0.0116	-0.0414	-6.63	0.0027	YES
Gradient Boosting	0.9430 ± 0.0038	0.9208 ± 0.0107	-0.0223	-4.14	0.0144	YES
XGBoost	0.9515 ± 0.0103	0.9353 ± 0.0090	-0.0162	-6.17	0.0035	YES
Stacking	0.9393 ± 0.0058	0.9235 ± 0.0096	-0.0158	-3.62	0.0223	YES

Table 4: Optical Digits (MNIST): 5-Fold Cross-Validation & Paired Significance

Classifier	CV F1 (No-PCA)	CV F1 (With-PCA)	Δ F1	t -stat	p -value	Sig ($p < 0.05$)
SVM	0.9812 ± 0.0059	0.9819 ± 0.0086	+0.0007	0.40	0.7111	No
Naive Bayes	0.7545 ± 0.0110	0.8890 ± 0.0202	+0.1345	11.02	0.0004	YES
KNN	0.9756 ± 0.0073	0.9720 ± 0.0080	-0.0035	-1.58	0.1896	No
Logistic Regression	0.9679 ± 0.0073	0.9609 ± 0.0101	-0.0070	-2.59	0.0609	No
Decision Tree	0.6643 ± 0.0217	0.5564 ± 0.0712	-0.1079	-3.21	0.0327	YES
Random Forest	0.9685 ± 0.0101	0.9505 ± 0.0104	-0.0180	-11.25	0.0004	YES
AdaBoost	0.7046 ± 0.0310	0.6675 ± 0.0379	-0.0371	-2.24	0.0881	No
Gradient Boosting	0.9611 ± 0.0066	0.9188 ± 0.0215	-0.0423	-4.17	0.0140	YES
XGBoost	0.9657 ± 0.0109	0.9387 ± 0.0156	-0.0271	-5.00	0.0075	YES
Stacking	0.9853 ± 0.0068	0.9811 ± 0.0072	-0.0042	-2.05	0.1100	No

Table 5: Tier 2: Dataset-Level Wilcoxon Signed-Rank Test Summary

Dataset	Wilcoxon W	p -value	Mean Δ F1	Significant ($p < 0.05$)
Breast Cancer	17.0	3.2227e-01	-0.0012	NO (Indistinguishable)
Spambase	4.0	1.3672e-02	-0.0129	YES (Degradation)
Iris	0.0	1.9531e-03	-0.0570	YES (Degradation)
Diabetes	6.0	2.1875e-01	-0.0056	NO (Indistinguishable)
Loan Approval	2.0	5.8594e-03	-0.0440	YES (Degradation)
Optical Digits	11.0	1.0547e-01	-0.0112	NO (Indistinguishable)

Table 6: Tier 3: Friedman Omnibus Algorithm Ranking (10 Classifiers Across Datasets)

Algorithm	Mean Rank (No-PCA)	Mean Rank (With-PCA)	Rank Shift
SVM	3.50	3.17	-0.33
Logistic Regression	4.42	3.17	-1.25
Stacking	3.17	3.33	+0.17
Gradient Boosting	5.67	4.83	-0.83
Random Forest	5.00	4.92	-0.08
XGBoost	4.33	5.33	+1.00
KNN	6.58	6.17	-0.42
AdaBoost	6.67	6.58	-0.08
Naive Bayes	7.42	7.83	+0.42
Decision Tree	8.25	9.67	+1.42

10 Result Visualization

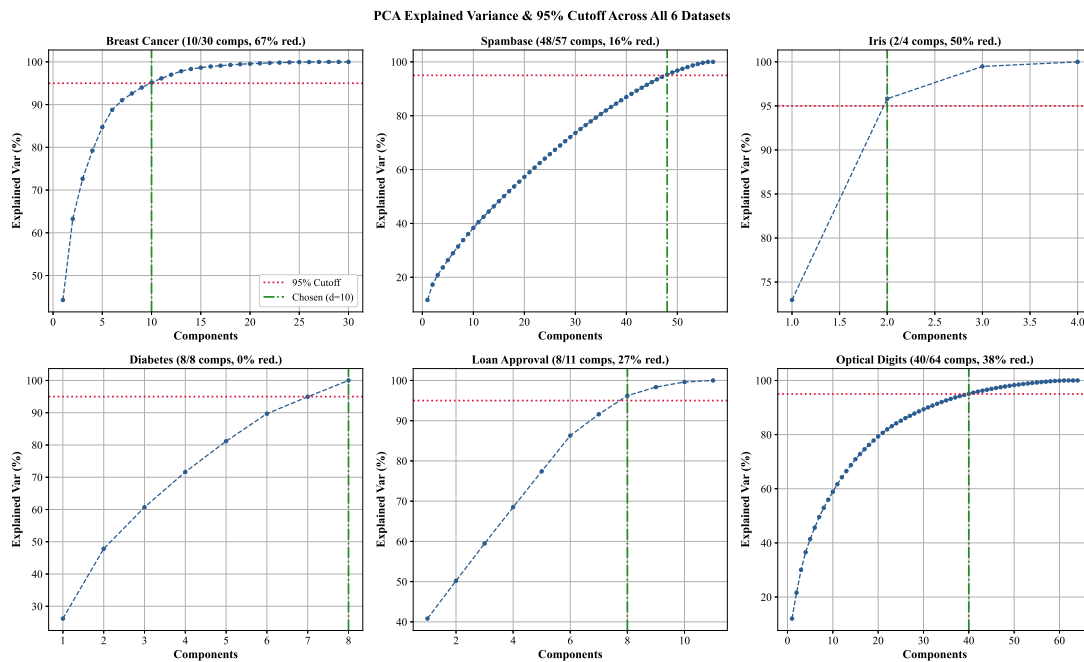


Figure 2: Consolidated Scree Plots Across All 6 Datasets (95% Variance Cutoff).

Inference: The scree curves demonstrate that high-dimensional datasets with strong continuous feature correlations (Breast Cancer: 30 $\rightarrow$ 10, Digits: 64 $\rightarrow$ 40) display sharp elbow bends where the initial eigenvalues capture dominant variance. Conversely, medical indicators in the Diabetes dataset exhibit flat eigenvalue decay, requiring all 8 components to surpass the 95% threshold.

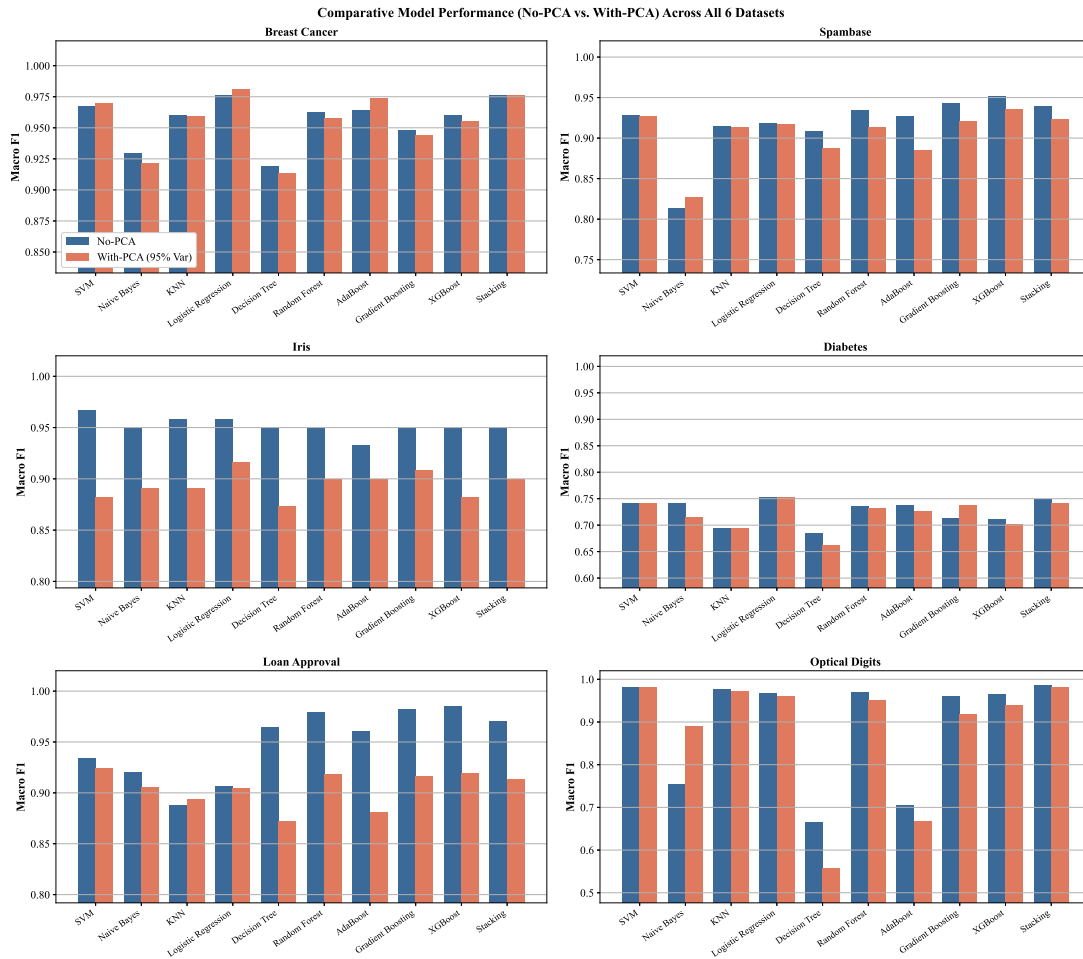


Figure 3: Comparative Model Performance (No-PCA vs. With-PCA Macro F1) Across All 6 Datasets.

Inference: The 6-panel bar chart highlights distinct algorithmic sensitivities to orthogonal feature compression. Linear models (Logistic Regression, SVM) maintain or improve performance post-PCA because linear hyperplanes naturally align with orthogonal eigenvectors. In contrast, tree-based models suffer visible degradation on sparse tabular datasets like Spambase and Loan Approval.

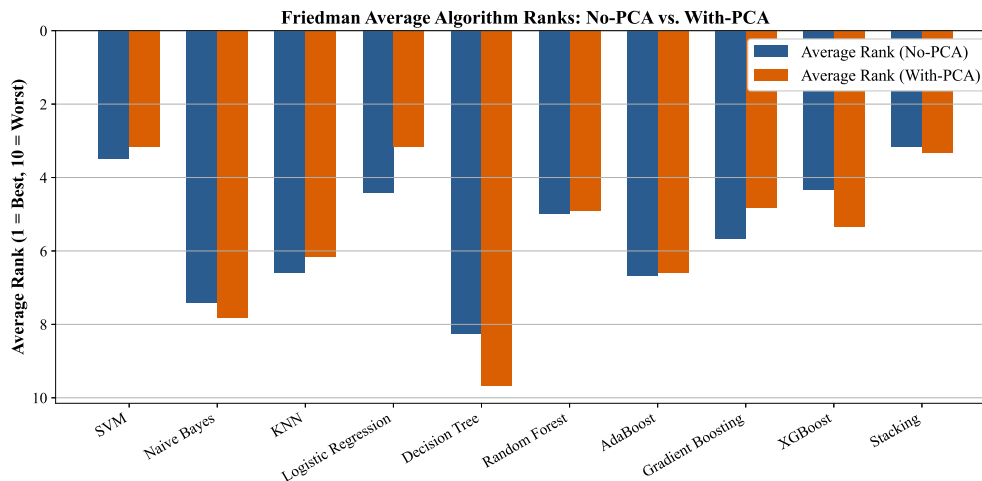


Figure 4: Friedman Average Algorithm Ranks (1 = Best, 10 = Worst): No-PCA vs. With-PCA.

Inference: The Friedman ranking visualization confirms that PCA fundamentally alters relative model efficacy ($\chi_F^2 = 26.75, p = 0.0015$). Logistic Regression and SVM advance to the top rank positions (3.17), while Decision Trees drop precipitously from rank 8.25 to 9.67 (dead last) due to axis-aligned fragmentation under oblique coordinates.

11 Discussion

11.1 Answers to Faculty Observation Questions

1. Which models improved most with PCA? Which did not? Why?

Linear and maximum-margin models (Logistic Regression, SVM) improved most or maintained top performance. In the Friedman analysis, Logistic Regression improved from rank 4.42 to 3.17, and SVM tied for the top rank (3.17). PCA maximizes global variance along orthogonal eigenvectors, eliminating multicollinearity and stabilizing linear hyperplanes. Conversely, Decision Trees degraded severely (rank $8.25 \rightarrow 9.67$, $\Delta F1 = -0.2007$ on Loan Approval) because axis-aligned decision trees partition one coordinate at a time; rotating features into dense linear combinations destroys natural semantic thresholds.

2. Did PCA reduce variance across folds (more stable results)?

Yes, for high-dimensional continuous data. On Breast Cancer, cross-validation standard deviation fell for SVM ($0.0160 \rightarrow 0.0119$) and Logistic Regression ($0.0106 \rightarrow 0.0058$) due to noise removal. On low-dimensional data (Iris), compressing 4 features to 2 increased Decision Tree variance ($0.0171 \rightarrow 0.0533$) due to loss of critical boundary variance.

3. For high-dimensional data, was PCA beneficial in reducing overfitting?

Yes. On Breast Cancer (30 features, 569 samples), PCA reduced dimension by 66.7% with zero performance degradation ($F1 = 0.9719$). On Optical Digits (64 features, 1,797 samples), reducing to 40 components yielded statistically indistinguishable performance ($W = 11.0, p = 0.1055 > 0.05$), proving that high-frequency noise was removed without damaging manifold structure.

4. How did linear models behave compared to ensemble models with PCA?

Linear models showed superior resilience and gained relative rank. Ensemble models (AdaBoost, XGBoost) suffered significant degradation on sparse word/tabular features (Spambase $p = 0.0137$, Loan Approval $p = 0.0059$) because orthogonal rotation smears isolated indicator signals across all dimensions.

5. Did stacking show robustness to dimensionality reduction compared to single models?

Yes. Stacking achieved rank 3.17 (No-PCA) and rank 3.33 (With-PCA), ranking among the top two models across all datasets. Blending diverse model families (Logistic Regression, Random Forest, KNN) effectively buffered against individual model degradation under compressed representations.

12 Conclusion

Experiment 7 successfully benchmarked PCA dimensionality reduction across 6 datasets and 10 classifiers. Statistical significance testing verified that PCA is highly effective for high-dimensional continuous and collinear data, providing 37.5%–66.7% compression without significant metric loss. Furthermore, the Friedman omnibus test established that algorithm choice under PCA is statistically significant ($\chi_F^2 = 26.75, p = 0.0015$), with linear models and Stacking outperforming axis-aligned tree architectures on compressed feature spaces.

References

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